

Azure Advisor Driven Reliability Improvement Sprint

ABSTRACT

The Azure Advisor Driven Reliability Improvement Sprint is a project that focuses on improving the reliability of applications and services running on Microsoft Azure. Azure Advisor provides recommendations when it finds possible issues in the cloud resources. In this project, these recommendations are studied and the important reliability issues are identified and prioritized. The team then works on fixing the selected issues and checks whether the changes have improved the system. The project also considers factors such as cost and dependency on resource owners. Overall, the project helps reduce the chances of service failures, improve system availability, and maintain a stable Azure environment.

PROBLEM STATEMENT

Cloud applications need to be reliable and available so that users can access services without frequent failures or interruptions. In Microsoft Azure, different issues such as incorrect configurations, lack of backup, or insufficient resource availability can affect the reliability of applications. **Azure Advisor** helps identify these issues by providing recommendations, but these recommendations need to be properly checked and acted upon.

This project focuses on finding the important reliability issues through Azure Advisor, understanding their impact, and fixing them based on priority. The team will implement the required changes and check whether the issues have been resolved successfully. While doing this, factors such as cost, available resources, and dependency on other teams or resource owners will also be considered. The main aim is to reduce the chances of failures and improve the overall stability and availability of the Azure environment.

USE CASES

1. **Identify Reliability Issues** – Check Azure Advisor to find problems or recommendations that may affect the reliability of Azure resources.
2. **Prioritize the Issues** – Review the identified issues and decide which ones need to be fixed first based on their impact and importance.
3. **Fix the Issues** – Make the required configuration or resource changes to solve the high-priority reliability problems.
4. **Check the Changes** – After fixing the issues, verify the Azure resources and Azure Advisor to make sure the problems have been resolved.
5. **Maintain Documentation** – Keep a record of the issues found, changes made, results, and any problems that are still pending.
6. **Consider Cost** – Check whether the proposed reliability improvements will increase the Azure cost and choose solutions that fit within the available budget.
7. **Handle Dependencies** – Identify issues that require help or approval from resource owners or other teams and keep track of their progress.

Requirements

Functional Requirements

1. The project should identify reliability issues using Azure Advisor.
2. The identified issues should be checked and given priority based on their importance.
3. The team should work on fixing the high-priority issues.
4. The status of each issue should be updated as pending, in progress, or completed.
5. After making changes, the team should check whether the issue is solved.
6. The work done and the results should be recorded for future reference.
7. Cost and dependency on other resource owners should be considered before making major changes.

Non Functional Requirements

1. **Reliability:** The Azure services should work smoothly with fewer failures.
2. **Security:** Azure resources should be managed safely.
3. **Performance:** The changes should not reduce the performance of the application.
4. **Scalability:** The solution should be useful for different Azure resources.
5. **Cost:** The improvements should be made without unnecessary increase in cost.
6. **Maintainability:** The changes should be easy to understand and maintain later.

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